NPLify · P0 Technical Baseline

Scenario Catalogue

New XP Technologies Limited

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Every scenario NPL runs today, grouped by client and project: 27 scenarios across 6 client groups. Each scenario is given as NPL described it (the story), the fee and the operational flow it uses, and the parts of the model that carry it. The ten operational flows the scenarios share are defined once in Part 2 and referenced by number (F1–F10); Part 3 lists the guided walkthroughs that trace a deal through the model entity by entity, as they appear on the interactive map; Part 4 walks one deal through the model class by class (X1–X7).

How to read this pack

Four map pages are now part of the pack as documents of their own, and are the place to start: the Top-Down Map (from the business, through the data model and the deal states, to the scenarios and the reading order), the Deal Flow on one page (Quote, Collect, Convert, Settle, with the roles and the money path), the Workflow Tree (8 classes, 27 sub-classes, 92 procedures, each naming who initiates, who approves, the codes it is written under and the records it touches) and the Workflow Tree on one page. Read them in that order, then this document. Every worked transaction (T), acceptance vector (L, B), walkthrough (W, X), flow (F) and procedure (C, A, D) has one identifier, listed in the Reading Companion's identifier register; the Test Strategy & Coverage document says which test proves each procedure.

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Part 1 · Scenarios by client

Client group	Projects	Scenarios
Betfair (BF)	BF, BF Sub, BF Withdrawal — three PROJECTs of one `CLIENT`; client-level balances and earnings from `CLIENT_POSITION`	6
Evo	Evo — the USDT → EUR invoice flow, plus Evo's bank-transfer and older currency products (dormant: not run for years, kept in scope by NPL)	9
NPL-GR (Game Reseller)	Game Reseller — NPL's own business unit as the client, run as an own-desk project	3
YN Remit, 66 Group, GDC, Peru (invoice-payment projects)	one `PROJECT` each; 66 Group is one `CLIENT` for its remittance and its game-reseller business	4
JTN, Jonas (cash projects)	JTN-JPY cash; Ad hoc INR cash	2
AK Personal	AK collections (USDT, INR)	3

The fees, rebates and rates quoted in the scenarios below are NPL's figures of 8 to 9 October 2026 and are illustrative: each is a configured value of the project (a FEE_STRUCTURE, FEE_OVERRIDE, MARKUP_SHARE_RULE or PARTNER_REBATE_RULE row, versioned by effective date), never a constant in code. In particular the Jeton rebate of 0.4 % (scenarios 1.1, 1.6, 2.5) is set per project (NPL, 20 October), and the figures of the fee table may differ from the values in force when a project goes live.

1 · Betfair (BF)

Projects: BF, BF Sub, BF Withdrawal — three PROJECTs of one `CLIENT`; client-level balances and earnings from `CLIENT_POSITION`

> Betfair customers fund their accounts in USDT converted to EUR; three partners, three fee mechanics, one withdrawal route back. Aquanow is used for everything today, Jeton is inactive, Ali is the alternative.

1.1 BF via Jeton: market minus fee, rebate back

Sender (Seven, Arrow Wood, Trade Lab, Orbit, Blue Sapphire) pays 1 % off market; Jeton keeps the full 1 % and pays NPL 0.4 % back monthly; Novi pays 0.9 % with 0.3 % back

FEE AND FLOW Fixed fee on market basis; F2 in, F5 out to BF's Sporting Exchange account

IN THE MODEL `FEE_STRUCTURE` (rate_basis market); `FEE_OVERRIDE` for Novi; `PARTNER_REBATE_RULE` 0.4 % of amount, scoped 0.3 % for Novi; `PARTNER_REBATE_ACCRUAL` reconciled against Jeton's statement; income in EARN_REBATE

1.2 BF via Ali: agent rate, customer-facing 1 %

Ali's 1 % sits inside its rate, NPL adds 0.4 %, the customer's calculation shows 'fee 1 %' on a recomputed source rate; Ali sometimes quotes 1.1738 the other way round

FEE AND FLOW Fixed fee on agent-rate basis; F2 in, F4/F5 by SWIFT under PT Sukses, F9 cutoff on the EUR settlement

IN THE MODEL customer_facing_fee_basis fixed 1 % on `FEE_STRUCTURE`, customer_facing_rate on `DEAL_GROUP`; quoted_rate and quoted_direction on `PARTNER_RATE_VERSION`; `QUOTE_PACKAGE` keeps the printed figures

1.3 BF via Aquanow: flat 0.8 %, rebate 0.3 %

Aquanow charges 0.8 % and pays NPL 0.3 % back monthly; EUR reaches BF by internal transfer, instantly

FEE AND FLOW Fixed fee; F2 in, F10 out

IN THE MODEL `PARTNER_PAIR` fee_on_market with partner_fee_pct 0.8 %; `PARTNER_REBATE_RULE` 0.3 % of amount; `RECEIVING_ENTITY_ACCOUNT`.rail internal

1.4 BF Sub via Aquanow or Ali: markup shared with Raeen

Downlines fund a sub-account Raeen provides at 4 %; the 4 % less the partner fee (3.2 % via Aquanow, 3 % via Ali) is split equally with Raeen; the EUR lands in Raeen's own Aquanow account, Raeen keeps its share and pays NPL monthly; Aquanow's 0.3 % rebate is NPL's alone

FEE AND FLOW Fixed 4 % with a share rule; F2 in, F10 out (Aquanow) or F5 under Raeen's name (Ali)

IN THE MODEL `MARKUP_SHARE_RULE` basis share_of_net_markup, party_kind client_party, settlement_mode party_retains, scoped by fee structure; `MARKUP_SHARE_ACCRUAL` as a receivable in SHARE_RECEIVABLE; rebate outside the base; `SETTLEMENT_SENDING_ENTITY` Raeen on the Ali route

1.5 Raeen direct collections via Aquanow (not sub-accounts)

Raeen collects from its downlines into its own BF account at 1 %; Aquanow's cost is 0.8 %; the 0.2 % left is split 75/25 — 0.15 % to Raeen, 0.05 % to NPL — plus Aquanow's 0.3 % rebate to NPL

FEE AND FLOW Fixed 1 % with a 75/25 share rule; F2 in, F10 out

IN THE MODEL A second `MARKUP_SHARE_RULE` scoped to this fee structure (Raeen 75 %); same party, same settlement by Raeen monthly

1.6 BF Withdrawal via Jeton

BF wires the customer's balance to Jeton's SWIFT account; Jeton converts at 1 % over market and pays the customer's TRC-20 or ERC-20 wallet; 0.4 % back monthly; Jeton is the only partner able to receive it

FEE AND FLOW Fixed 1 %; F4 in (bank at Jeton), F3 then wallet payout

IN THE MODEL Bank endpoint with rail; `RECEIVING_ENTITY_ACCOUNT` kind wallet; `DESTINATION_APPROVAL` with network and address; `WALLET_SCREENING` on the receiver wallet; rebate as above

2 · Evo

Projects: Evo — the USDT → EUR invoice flow, plus Evo's bank-transfer and older currency products (dormant: not run for years, kept in scope by NPL)

> Evo's customers pay their invoices to Evo through NPL; everything converts into EUR; two receiver groups with senders designated per group; Ali and Jeton can each pay only some entities. Evo's bank-transfer and older currency products have not run for years but stay in scope: the fee still applies because the currency is collected and then settled separately.

2.1 Standard USDT collection

Sender pays 1.4 % off the rate, Evo pays 0.6 % from its balance at conversion; sender sees only the sender rate

FEE AND FLOW 2 % split 1.4 / 0.6; F2 in (screening not done for every Evo sender), F5 out by SWIFT under Bplay

IN THE MODEL `FEE_STRUCTURE` sender_pays_pct / receiver_pays_pct; `CONVERSION`.fee_receiver_part; `COLLECTION`.screening_skipped

2.2 Selected senders (GTSI, HG, Komodo)

Sender pays 1.0 % while Evo still pays 0.6 %

FEE AND FLOW 1.6 % with the receiver part fixed

IN THE MODEL `FEE_OVERRIDE` with its own sender_pays_pct and receiver_pays_pct; `DEAL`.fee_override_id

2.3 Two receiver groups

Bplay group: Babylon, paid on an invoice reference; Bplay and Lex, self-transfers with an internal invoice or statement reference. TBet group: TBet, self-transfers with an internal invoice or statement reference. Ali pays Babylon and TBet; Jeton pays Bplay, Lex and TBet

FEE AND FLOW Configuration

IN THE MODEL `RECEIVER_GROUP`, `SENDER_RECEIVER_ALLOW` per sender, `DESTINATION_APPROVAL` per entity per partner; `SETTLEMENT_LINE`.reference carries the invoice or statement reference; the ledger rules

2.4 Collect-first and quote-first senders

Some pay what they can, others the exact invoice

FEE AND FLOW Entry mode per sender

IN THE MODEL `DEAL_GROUP`.entry_mode collect_first / quote_first

2.5 Via Jeton instead of Ali

Jeton charges 1.25 % overall and pays 0.4 % back monthly; Jeton collects USDT only

FEE AND FLOW F2 in, F5 out

IN THE MODEL `PARTNER_PAIR` fee_on_market with partner_fee_pct 1.25 %; `PARTNER_REBATE_RULE` 0.4 % of amount

2.6 Other currencies via Ali, and the 14:30 cutoff

Rate taken from Ali in the morning and compared; a deal locked before 14:30 GMT+8 converts the same day, after it the collection goes ahead on the morning's figures and converts the next business day; NPL absorbs the difference; an unacceptable next-day rate is renegotiated or deferred a day

FEE AND FLOW F4 in, F9

IN THE MODEL `PARTNER`.bank_cutoff_time, `PARTNER_CUTOFF_OVERRIDE`, `HOLIDAY`; `DEAL`.rate_locked_at, cutoff_at, conversion_due_date, conversion_deferred_reason; cutoff_timing variance; pending value in `PROJECT_BALANCE`

2.7 Returned payout

A bank rejects the transfer

FEE AND FLOW F6

IN THE MODEL `SETTLEMENT_RETURN`, reversing posting, return leg to the sender_return receiver

2.8 Same-currency pass-through (USD, EUR, GBP) — dormant

A bank transfer in and the same currency out; 1.4 % from the sender and 1.4 % from Evolution

FEE AND FLOW 2.8 %; F4 in, F5 out, F9

IN THE MODEL `CURRENCY_PAIR` with equal sides, `CONVERSION` at rate 1; `FEE_STRUCTURE` sender_pays_pct 1.4 / receiver_pays_pct 1.4

2.9 CNY, PHP, THB, SGD, HKD; AED and USD cash in Dubai (older currency pairs) — dormant

Older currency pairs of the same client at 3.2-6.75 %; AED cash rounds to the nearest 100, USD cash in Dubai to the nearest 10

FEE AND FLOW F4 or F1 in

IN THE MODEL Standard pairs; `CURRENCY_PAIR`.cash_rounding nearest_n with cash_rounding_unit 100 for AED and 10 for USD cash

3 · NPL-GR (Game Reseller)

Projects: Game Reseller — NPL's own business unit as the client, run as an own-desk project

> NPL resells games at retail against wholesale vendor invoices and collects its receivables to pay the vendors; Management confirmed running it inside NPLify as an own-desk project.

3.1 USDT customers (Bala, Fair, 66 Group GR, TripleSeven, Alpha Plus)

Customer owes EUR, pays USDT straight into NPL's own wallet (LT Sub at Aquanow, occasionally Ali's wallet as NPL's own balance); 2 % markup over market; the USDT collected is rounded up

FEE AND FLOW Fixed 2 % (0.75-1.5 % for some); F2 in to NPL's wallet

IN THE MODEL `CLIENT` NPL-GR with its `PROJECT`; `PARTNER_CONFIG`.is_own_desk; `OWN_WALLET` as `COLLECTION_RECEIVING_ENDPOINT`; the ledger rules restated; `CURRENCY_PAIR` round_up_unit

3.2 27 Group: THB cash

Customer pays THB cash by token through Ali at market + 2 %; Ali converts THB to USDT, which moves to NPL's wallet or stays at Ali as NPL's balance; Ali's rate must stay within 1.5 % of market so NPL keeps 0.5 %

FEE AND FLOW Fixed 2 %; F1 in; INR cash possible later by the same process

IN THE MODEL `DEAL`.cash_location, `COLLECTION`.token_ref, counted_at; `PARTNER_RATE_VERSION` breach threshold 1.5 % warning Finance; min_margin_pct 0.5 %

3.3 Vendor invoices paid in parts

Customers pay against a EUR invoice or an advance; vendors (SACCO SIA, Mitratech, EZ-EVO, BraversPlay) are paid in USDT on different days at that day's rate; the forex difference is accepted within the 2 %

FEE AND FLOW USDT settlements to vendor wallets, F3 before each

IN THE MODEL `INVOICE` under `PROJECT`; `DEAL_GROUP`.invoice_id; `SETTLEMENT_LINE`.obligation_discharged, discharge_rate, discharge_market_rate_id; `INVOICE_BALANCE` view; fx_timing variance

4 · YN Remit, 66 Group, GDC, Peru (invoice-payment projects)

Projects: one `PROJECT` each; 66 Group is one `CLIENT` for its remittance and its game-reseller business

> A client sends USDT (or a bank transfer) to pay a third party's invoice by bank transfer; quote-first, priced at the partner's rate plus NPL's markup.

4.1 YN Remit

Third-party invoices with a clear sending and receiving entity and KYC onboarded on Ali's PT channels; quote first at partner rate + 0.5 %; USDT to USD, EUR or GBP; occasionally NPL's own entity is the declared sender

FEE AND FLOW Variable 0.5-2.19 %; F2 in, F4/F5 out, F6 if returned, F9

IN THE MODEL Quote-first `DEAL_GROUP`; `DESTINATION_APPROVAL` per route; `SETTLEMENT_SENDING_ENTITY` may be NPL's entity; `FEE_STRUCTURE` mode variable

4.2 66 Group, multi-currency and INR cash

Like YN Remit; INR almost always collected as cash at 7 % total (average 6 %, minimum margin 0.5 %), rounded to 1,000; a local INR account is arranged on demand for one transaction

FEE AND FLOW Variable, set per deal; F1 in (cash) or F7 (one-time local account), F5 out under NPL's name via PT Global

IN THE MODEL `FEE_STRUCTURE` variable with min_markup_pct, max_markup_pct, min_margin_pct; cash_rounding to 1,000; `DEAL`.cash_location, token; `COLLECTION_RECEIVING_ENDPOINT` usage single_use and `LOSS_EVENT` if an account is frozen

4.3 GDC, fee by monthly volume

USDT to USD at 1: 1 plus 2.5 %, 2.25 % once the calendar-month volume passes USD 500,000, the crossing deal split; several receiving entities (Adu Ads, Ad Fusion, TDC, Pixel Labs); New Pinnacle Ltd contracts; minimum margin 1.8 %

FEE AND FLOW Tiered; F2 in, F5 out via PT Global, F9

IN THE MODEL `FEE_TIER` with `PROJECT_MONTH_VOLUME`; `DEAL_FEE_TIER` portions, blended fee; one `RECEIVER` with several RECEIVING_ENTITY_ACCOUNTs; `SETTLEMENT_SENDING_ENTITY` New Pinnacle Ltd

4.4 Peru remittance

Bank transfer to Ali's receiving entity PT Silver Lining Production, converted to USDT; Ali 3 % + NPL 1 %

FEE AND FLOW Fixed 4 %; F4 in, F3 then wallet payout

IN THE MODEL `PARTNER_PAIR` fee_in_rate; bank-in endpoint with rail

5 · JTN, Jonas (cash projects)

Projects: JTN-JPY cash; Ad hoc INR cash

> USDT in, local cash out, or local cash in, USDT or USD out; the same party on both sides, priced per city, settled by token.

5.1 JTN-JPY cash

JTN sends USDT to Ali; Ali converts to JPY cash handed over by token; the day's rate is compared and quoted with a markup of 0.2-0.5 %, maximum 0.5 %; total fee 1-1.8 %

FEE AND FLOW Variable with a maximum markup; F2 in, F1 out

IN THE MODEL `FEE_STRUCTURE` mode variable with max_markup_pct 0.5 %; sender equals receiver; `SETTLEMENT_LINE`.token_ref; `DEAL`.cash_location

5.2 Ad hoc INR cash (Jonas)

INR cash collected by token in a named city at 7 % total, paid out in USD, SGD or USDT; rates differ between Delhi and Mumbai

FEE AND FLOW Fixed 7 %; F1 in, wallet or F5 out

IN THE MODEL `DEAL`.cash_location, `PARTNER_RATE_VERSION`.location, `COLLECTION`.token_ref, counted_at; cash_rounding to 1,000

6 · AK Personal

Projects: AK collections (USDT, INR)

> A stakeholder's account operated with Ali: collections and conversions as normal, but the balance stays at the partner and is paid out on request, with no markup.

6.1 Held balance

USDT (and INR) collected; balances kept in USD, SGD and USDT up to about USD 400,000 equivalent, revised from time to time

FEE AND FLOW 0 %; F2 in

IN THE MODEL `FEE_STRUCTURE` 0 % with min_margin 0 %; `PROJECT`.settlement_policy hold_allowed, exposure_ceiling; `PROJECT_BALANCE` view (per project), `CLIENT_POSITION` across the client's projects

6.2 Conversions between held currencies

10,000 USDT collected at night converts to SGD the next business morning: USD and SGD are banking-ready balances bound by the cutoff

FEE AND FLOW F9

IN THE MODEL `BALANCE_CONVERSION` with rate_locked_at and conversion_due_date; `ENTITLEMENT_REATTRIBUTION`

6.3 Withdrawals on request

Local bank payment, SWIFT, USDT or cash

FEE AND FLOW F8 (local), F5 (SWIFT), F3 then wallet, F1 (cash)

IN THE MODEL `SETTLEMENT` with the matching rail; `SETTLEMENT_LINE`.token_ref for cash; `RECEIVING_ENTITY_ACCOUNT` single_use for a one-time local account

Workflow names (awaiting NPL)

Pete asked for more intuitive names for several workflows without saying which. The renaming waits for NPL's list. When it arrives, the names change in the Workflow Tree, the common procedures (C1 to C10), the state tables of the Lifecycle & Approvals document and this catalogue, and the old names are kept in the glossary (ERD Appendix A.1) with a pointer to the new ones, so that older references still resolve. Nothing else in the pack depends on a workflow's name: codes and identifiers are unchanged by a rename.

Part 2 · Shared operational flows

Defined once and referenced by number from the scenarios above. The step-by-step version of each, with who does what and the evidence recorded, is in the Operating Procedures (common procedures C1–C10).

F1 Cash by token (collection or settlement)

Collection: the partner issues a token (a local-currency bill) and the agent's contact; the sender meets the agent, hands over the cash, keeps the token and shares its photo; the partner counts and confirms. The rate is the one locked at quote even if the hand-over or the count slips to the next day; the partner may replace the agent and the token before the meeting, and NPL passes the new details to the sender. Settlement: the receiver provides the token up front to NPL, NPL to the partner, the partner to the agent; the agent takes the token, hands over the cash, photographs the token for the partner, who passes it to NPL and NPL to the receiver. A cash-in deal paid out in USDT has no cutoff; one paid out by bank does

In the model `COLLECTION`.token_ref (replaceable before hand-over), agent_contact_ref, evidence_ref (the photo), counted_at, states token_issued, handed_over, counted, verified; `DEAL`.cash_location; `PARTNER_RATE_VERSION`.location; `SETTLEMENT_LINE`.token_ref (the receiver's token) and the agent's photo as evidence

F2 USDT collection with wallet screening

Sender declares a wallet; NPL screens it; standard or alternate receiving wallet by risk; sender shares the hash; NPL verifies on chain; partner confirms. Where a sender cannot share a wallet up front (Evo and other projects) NPL arranges an alternative address: each partner has designated wallets for such cases, and where the partner does not support the network (Ali has no ERC-20) NPL collects on its own alternate wallet on that network and forwards to the partner on a network it supports (TRC-20), without the risk exposure

In the model `WALLET_SCREENING` (Operations); `COLLECTION`.sending_address, wallet_screening_id, screening_skipped; `COLLECTION_RECEIVING_ENDPOINT`.risk_band for the partner's designated wallets; `OWN_WALLET` as a bridge endpoint forwarded to the partner as a hop — a model amendment is proposed for this

F3 Receiver wallet screened before settlement

NPL screens the receiver's wallet, the partner executes, the hash is shared, the receiver confirms

In the model `WALLET_SCREENING` subject receiver_wallet; `DESTINATION_APPROVAL`.wallet_screening_id

F4 SWIFT collection

Partner approves the sender from company profile and KYC and provides an account; proof of transfer exchanged; partner confirms; conversion per the agreed flow

In the model `COLLECTION_SENDING_ENTITY`; bank endpoint active only with its rail and matching the leg; `COLLECTION`.evidence_ref, verified_by, verified_at

F5 SWIFT settlement

Partner approves the destination and route before collection; transfer after collection; slip shared with the receiver; receiver confirms; on special cases the client asks for the bank fee to be covered — Finance reviews and recommends, Management approves; an unreceived transfer is chased through the banking channel

In the model `DESTINATION_APPROVAL`; `SETTLEMENT`, `SETTLEMENT_RAIL`, `SETTLEMENT_SENDING_ENTITY`; `CONFIRMATION`; `BANK_FEE_EVENT` with a `FEE_DECISION` recommended by Finance and approved by Management

F6 Returned transfer

Resend, reuse the returned funds for another transfer, or reconvert and return to the sender

In the model SETTLEMENT_RETURN; reversing posting; return leg to the sender_return receiver

F7 One-time local deposit account

The partner provides an ad hoc account for one transaction and amount; deposit or local transfer; proof exchanged; partner recognises the funds; a frozen account caused on the sender's side is a reversal of the collection; a loss can be partial or shared

In the model COLLECTION_RECEIVING_ENDPOINT usage single_use, reserved_for_deal_id, expected_amount; LOSS_EVENT with the four-way split, EXP_LOSS; COLLECTION state reversed

F8 Local payouts for small amounts

Receiver gives a local account where the partner allows it (SG, HK, Indonesia, Thailand); the partner pays by cash deposit into that account or by local transfer and shares proof; receiver confirms. A cash deposit by the agent into the receiver's bank account is a local bank payment, not a cash hand-over: no token. Cash short at hand-over is adjusted later or paid separately

In the model RECEIVING_ENTITY_ACCOUNT usage single_use; SHORTFALL topped up next settlement or by its own SETTLEMENT; LOSS_EVENT if unrecoverable

F9 Bank cutoff

Per partner where the partner has one — Ali and other Asian partners; Jeton and Aquanow have none, so their cutoff is empty and nothing is bound. For pairs with a bank side only: binds the rate lock and the outgoing settlement; locked before 14:30 GMT+8 is same day, after it next business day at that day's rate (unless locked within the previous day's cutoff); extendable 30-60 minutes if the partner agrees; closed early on Fridays or before a holiday

In the model PARTNER.bank_cutoff_time (nullable: no cutoff for Jeton, Aquanow), PARTNER_CUTOFF_OVERRIDE, HOLIDAY; PARTNER_PAIR.needs_bank_lock; DEAL and SETTLEMENT cutoff_at and due dates; cutoff_timing variance

F10 Internal transfer inside the partner's platform

Aquanow moves EUR to BF's account on the same platform, instantly

In the model RECEIVING_ENTITY_ACCOUNT.rail internal; no bank fee on the rail

Part 3 · Guided walkthroughs

Each walkthrough lights up the path a deal takes on the interactive map, one entity at a time. They are listed here so the catalogue is complete on paper.

W1 · Quote-first deal, end to end (20 steps)

Entity	What happens there
PROJECT	One arrangement with the client (CLIENT → PROJECT). Its policies (quote freshness and validity, grace, fee treatment, settlement policy and exposure cap) govern everything below.
SENDER	Operator picks the sender. The receiver groups this sender may pay are listed in SENDER_RECEIVER_ALLOW.
RECEIVER_GROUP	The entitled group. One of its receivers' accounts must hold an approved destination (DESTINATION_APPROVAL) with the chosen partner before the deal leaves Inquiry.
PARTNER_PAIR	Choosing partner × currency pair is part of pricing. Partner and partner entity are fixed per leg.
FEE_STRUCTURE	The versioned client-side fee for that partner × pair is stamped on the deal group.
RATE_COMPARISON	Created lazily for partner × pair × date by the first deal of the day.
MARKET_RATE	The market snapshot from the pair's prioritised rate source.
PARTNER_RATE_VERSION	The partner's rate version for the day; the breach check may freeze the pair until Finance decides.
DEAL_GROUP	The client-facing instruction: pair as the sender sees it, sender rate, fee, reference. Every deal belongs to one; this one has a single leg.
DEAL	Leg 1, forward, funded by a collection. Priced once; sent / valid / grace timestamps and quote_hash recorded. A sender who asks for more time before expiry gets Finance-approved extensions, as many as needed, at the same price; after expiry a new deal group, which may be priced on the old rate by an approved override.
QUOTE_PACKAGE	The rendered quote as sent: template version, channel, printed sender-facing figures, sent_at (starts validity) and the hash that becomes the leg's quote_hash. A re-issue is a new row; a new price is a new deal group.
DESTINATION_APPROVAL	Destination approval precedes collection: an approved account of the entitled group with this partner.
COLLECTION	Funds arrive in 1..n parts into a COLLECTION_RECEIVING_ENDPOINT of the leg's partner entity (bank or wallet by kind); cash with instructions on the deal.
CONVERSION	The partner converts against the exact rate version. actual_out is booked rounded as the pair says; a dropped fraction stays as client dust, a round-up is NPL's cost; the receiver fee is taken here; NPL's margin is captured now and never again at risk.
LEDGER_TRANSACTION	One balanced transaction; margin recognition is the only client → company ownership flip.
SETTLEMENT	Pays the balance, rounded as the pair says, from the partner's holding through a SETTLEMENT_RAIL with a SETTLEMENT_SENDING_ENTITY narrative.
SETTLEMENT_LINE	One or many destination accounts. Paying a receiver in another group creates the offset in the same action.

Entity	What happens there
CONFIRMATION	The receiver confirms the amount received: full or short.
SHORTFALL	If short, a shortfall opens against the receiver and is topped up by a later settlement.
BANK_FEE_EVENT	Post-confirmation true-up of expected vs actual bank fee: waived, carried forward or absorbed.

W2 · Collect-first deal (6 steps)

Entity	What happens there
COLLECTION	Funds arrive before any quote. Attribution needs an existing SENDER_RECEIVER_ALLOW entry; the sender is picked from the configured list.
DEAL_GROUP	Created with entry_mode collect_first; the sender is promised nothing.
DEAL	The single leg is born in state Collected with priced_at = conversion time; no quote clocks.
PARTNER_RATE_VERSION	The rate version is taken at conversion rather than stamped at quote.
CONVERSION	Converted against that version; actual_out booked; margin captured.
SETTLEMENT	Then the same whole-unit settlement as any other deal.

W3 · Rejected payout and return leg (10 steps)

Entity	What happens there
SETTLEMENT	USDT was converted to GBP at PT and paid out to the receiver's account.
SETTLEMENT_RETURN	The receiving or intermediary bank rejects the transfer. The GBP returns to PT minus bank fees; the line's outcome becomes returned and a reversing transaction puts the money back into the client's balance.
BANK_FEE_EVENT	The bounce fee is booked as expected vs actual, waived (client) or absorbed (NPL) per the project's treatment.
GROUP_ENTITLEMENT	With the line reversed, the group's entitlement is restored. Option B stops here: the GBP stays in balance for later payouts to receivers in the same group; another group would be an approved offset.
DEAL	Option A: a return leg on the same deal group. leg_type return, funding_source returned_settlement, GBP → USDT at PT, priced at conversion. NPL's original margin is untouched; the FX loss falls on the client's funds.
FEE_DECISION	Markup on the return leg is applied or waived case by case, with an approver.
CONVERSION	The reverse conversion, owned by the return leg, using PT's rate version for GBP → USDT.
RECEIVER	The destination is the sender_return receiver linked to the parent leg's sender. Any other receiver needs Management approval, a reason, and a prior destination approval.
RECEIVING_ENTITY_ACCOUNT	The sender's USDT wallet is one of that receiver's accounts; it is chosen at payout and need not be the wallet the sender paid from.
SETTLEMENT_LINE	The USDT goes out as an ordinary payout line to that account, clearing the entitlement its return leg credited to the parent leg's group.

W4 · Two-leg route via a deal group (9 steps)

Entity	What happens there
DEAL_GROUP	The sender wants INR → EUR. Ali can collect INR but cannot reach EUR, so the group will have two legs. The group holds the one price the sender saw.
DEAL	Leg 1: INR → USDT at Ali, forward, funded by the collection. Partner fixed per leg.
CONVERSION	Ali converts to USDT; margin on leg 1 is captured here.
RECEIVER	Jeton's collection wallet is registered as a RECEIVER of kind partner_transit, so the hop is an ordinary payout.
SETTLEMENT	Ali pays the USDT to the transit receiver's account. Entitlement passes through the transit group.
PARTNER_CONFIG	Leg 2 is a second DEAL on the same group at Jeton: USDT → EUR, funding_source prior_leg, no collection.
PARTNER_RATE_VERSION	Jeton's rate version for USDT → EUR prices leg 2 at conversion.
SETTLEMENT_LINE	Jeton pays the EUR to the real receiver's account. Margin across the route = group quote – both partner conversions.
GROUP_ENTITLEMENT	The entitled group's tally reconciles once leg 2 converts and pays out.

W5 · Ad-hoc balance conversion (8 steps)

Entity	What happens there
PROJECT_BALANCE	The client holds 5K SGD and 10K USDT at PT and must pay 15K USDT.
BALANCE_CONVERSION	Finance converts SGD → USDT at PT. Not a deal: no sender, receiver or collection. It names the group it serves (chosen split).
PARTNER_RATE_VERSION	The same rate procedure applies: partner rate version, market snapshot, breach check.
FEE_DECISION	Markup is optional, decided case by case with an approver.
CONVERSION	Owned by the balance conversion; actual_out rounded as the pair says, the residual kept as dust.
ENTITLEMENT_REATTRIBUTION	The served group's SGD entitlement becomes USDT entitlement. Any excess over its SGD entitlement is another group's flow, booked as an approved offset.
LEDGER_TRANSACTION	Source BalanceConversion: debit client due USDT, credit client due SGD; an earnings posting only if markup was applied.
SETTLEMENT	The 15K USDT goes out as an ordinary settlement.

W6 · Reroute, custody and recover dues (6 steps)

Entity	What happens there
DEAL	A priced leg hits a partner problem: unavailable, limit reached, holiday, compliance hold.
REROUTE	The only way partner_pair / facilitating_entity change after quote. Pre- or post-collection; post-collection outcome is via the own wallet or a refund to the sender. Approved by Finance.
OWN_WALLET	Contingency custody. CL.OWN.*.HELD may only be debited by a collection on a deal with an approved reroute; custody_max_hours bounds it.

Entity	What happens there
PARTNER_CONFIG	The alternate partner must have applicable_for_reroute set. NPL's cut is captured at whichever partner converts, so nothing is withheld.
EARNINGS_RECEIVABLE	With forward_and_recover_dues, NPL forwards less than the full amount and reduces its receivable at that partner by the same figure: own wallet cash up, receivable down, no income. The partner's agreement is recorded.
CUSTODY	The custody aging view (collections under open reroutes plus recovered dues resting in the wallet, minus forwarded) is the standing control.

W7 · Rate day and breach approval (7 steps)

Entity	What happens there
RATE_SOURCE	Kraken, CMC, OANDA or XE, each with a staleness threshold.
PAIR_RATE_SOURCE	The prioritised source chain for each currency pair.
MARKET_RATE	Captured rate per source and pair; cross-source slippage is a query over these.
RATE_COMPARISON	One per partner × pair × date, with 2w and 4w average spreads built from confirmed versions only. A partner may give a new rate later the same day.
PARTNER_RATE_VERSION	Breached when spread > avg_2w × (1 + variance_threshold_rel). Quoting and conversion on that partner × pair freeze; Finance may also ask for a better rate.
APPROVAL	Finance approves with a reason or rejects. Approver ≠ initiator.
USER	Roles: ops, finance, management. Operations never sees derived-economics fields.

W8 · Ledger and balances (7 steps)

Entity	What happens there
CONVERSION	A source event (collection, conversion, balance conversion, settlement, return, confirmation, bank fee event, reroute, markup share, rebate, loss event, adjustment).
LEDGER_TRANSACTION	One transaction per event, append-only; a correction is a reversing transaction.
LEDGER_POSTING	Debit and credit lines that sum to zero per currency, tagged client or company and by cost component.
LEDGER_ACCOUNT	Accounts coded OWN.HOLDER.CCY.PURPOSE; client due balances are one account per project × partner × currency. ADVANCE no longer exists.
PROJECT_BALANCE	Conversions in, balance conversions across currencies, returns back in, settlements out. Never stored; carries dust below one unit.
GROUP_ENTITLEMENT	Per group and currency: forward legs' client_net ± re-attributions − paid + returned. Σ groups = Σ client balance.
OFFSET	Positive = owed, negative = holding another group's flow. Zero by construction in a single-group project.

Part 4 · Worked examples: one deal through the model, class by class

Pete asked for each key workflow to be supported by one example that traces the transaction from beginning to end through the ERD, showing how each entity is affected, with a ledger view of how the numbers move. The seven examples below follow the seven classes of the Workflow Tree that carry figures (class 1, Configure & set up, carries none). They reuse the figures of the pack: project Evo, Sender A, Receiver X (entities X1, X2) in the default group, Ali converting at 0.8910 against market 0.9000, fee 2.00 % split 1.40 % sender and 0.60 % receiver; the T numbers are the Ledger Posting Design's and the L numbers the Calculation Specification's, so every figure here can be checked there. The same examples become test fixtures in Phase 1 (M2 to M5).

X1 · Rates and quote (classes 2 and 3): deal 1 is priced and sent

A quote-first deal of 10,000 USDT for Receiver X.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Ali's rate of the day	PARTNER_RATE_VERSION : quoted 0.8910 EUR per USDT, direction as the pair; RATE_COMPARISON for Ali × USDT→EUR today; MARKET_RATE 0.9000 from Kraken	spread = 1 − 0.8910 ÷ 0.9000 = 1.00 %; two-week average 1.00 %, threshold 20 %: no breach (§2); Finance confirms the version	none	—
2. Price the inquiry	DEAL_GROUP (project Evo, Sender A, group DEFAULT, entry_mode quote_first, fee_structure_id stamped); DEAL leg 1 forward, priced_at now, partner_entity PT-tour; DEAL_FEE_TIER none (Evo has no tiers)	principal 10,000 × 0.9000 = 9,000.00; sender part 1.40 % = 126.00; receiver part 0.60 % = 54.00; sender rate 0.88740; gross out 8,874.00; client net 8,820.00 (§3, vector L1)	none: nothing posts at pricing	—
3. Send the package	QUOTE_PACKAGE issue 1 with sent_at and quote_hash; validity 60 minutes from sent_at (PROJECT)	the package prints the sender-facing figures only: sender rate 0.88740, customer-facing fee 2.00 %, amount out 8,874.00, valid until	none	—
4. Sender accepts; wallet screened; instructions issued	WALLET_SCREENING accept on Sender A's address; COLLECTION pending on PT-tour's TRC-20 endpoint; DEAL state Collecting	—	none	—

Pricing fields are now write-once. If Ali's version had breached, the pair would be frozen and step 2 could not run (Figure 3); if Sender A had let the validity lapse, Section 3.2 of the tree applies (grace, then re-quote).

X2 · Collect (class 4): the 10,000 USDT arrive

Sender A pays and shares the hash.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Hash checked	COLLECTION: tx_hash, explorer result attached (address, amount, confirmations), partner confirmation; verified_by Operations, verified_at	10,000.00 USDT received = amount_in; no over- or underpayment	T1: CL.ALI.USDT.COLLECTED Dr 10,000.00 · CL.PROJECT.USDT.PAYABLE Cr 10,000.00 (receiver_group DEFAULT)	COLLECTED at Ali 10,000.00 USDT; PAYABLE 10,000.00 USDT; GROUP_ENTITLEMENT DEFAULT 10,000.00 USDT
2. Deal state	DEAL Collected; on a bank-channel pair also rate_locked_at and conversion_due_date (C9); here USDT→EUR at Ali needs_bank_lock, locked before 14:30 Hong Kong, so due today	—	none	unchanged

Had 10,050 arrived, the extra 50 would be a SENDER_CREDIT (T28); had 9,000 arrived, the deal would wait in Collecting for Finance to decide (A.3).

X3 · Convert (class 5): Ali converts

Operations instructs, Finance approves, Ali books.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Instruct	CONVERSION expected_out 8,910.00 (10,000 × 0.8910), calc_mode partner_calculates, state Expected; APPROVAL price/convert by Finance	expected_out = amount_in × partner rate (§12)	none	unchanged
2. Ali books	CONVERSION actual_out 8,910.00, variance 0.00, state Booked → Approved; fee_receiver_part 54.00 taken here (receiver_fee_timing at_conversion)	delivered 8,910.00; client net 8,820.00; NPL earnings 90.00 = delivered – client net (§8, L1)	T2: PAYABLE USDT Dr 10,000 · COLLECTED Cr 10,000 · CL.ALI.EUR.DUE Dr 8,820.00 · CO.ALI.EUR.POOL Dr 90.00 · CL.PROJECT.EUR.PAYABLE Cr 8,820.00 · CO.NONE.EUR.EARN_GROSS Cr 90.00	DUE at Ali 8,820.00 EUR (PROJECT_BALANCE); PAYABLE 8,820.00 EUR; POOL 90.00; EARN_GROSS 90.00; USDT jars 0

This is the only moment client money becomes NPL's. Through Jeton instead (fee on market 1.25 %) the same deal posts as T14: EXP_PARTNER 112.50, POOL 67.50, EARN_GROSS 180.00 (L2).

X4 · Settle (class 6): the euros are paid out

A run that also pays deal 2 and the EUR pass-through, 18,090.00 in all.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Prepare	SETTLEMENT (project Evo, rail "Ali's EU bank / EMI", narrative PT Global Inc, total 18,090.00); SETTLEMENT_LINE X1 8,000.00 and X2 10,090.00, each with deal_id and an approved DESTINATION_APPROVAL	amount_rounding exact: cents go out (§9)	none	unchanged
2. Approve and release	APPROVAL by Finance (below the 50,000 EUR settlement limit, first slot of the day); Ali releases; SETTLEMENT released_at; lines outcome sent, released_at, expected_confirmation_by = release + rail time	—	T7: CL.PROJECT.EUR.PAYABLE Dr 18,090.00 · CL.ALI.EUR.DUE Cr 18,090.00	DUE 0; PAYABLE 0; the two lines are "sent, not yet confirmed"
3. Confirm	CONFIRMATION line 1 full; line 2 short, 10,040.00 received; SHORTFALL 50.00 open for Receiver X	bank fee 50.00; treatment carry_forward	T8 nothing; T9: CL.RECEIVER_X.EUR.SHORTFALL Dr 50.00 · PAYABLE Cr 50.00	SHORTFALL 50.00; PAYABLE 50.00
4. Top up at the next run	BANK_FEE_EVENT applied; Management approves the top-up from the pool	NPL bears 50.00	T10: EXP_BANKFEE Dr 50.00 · POOL Cr 50.00 · DUE Dr 50.00 · SHORTFALL Cr 50.00	DUE 50.00 (out with the next run); POOL 225.00 after the three deals; EXP_BANKFEE 50.00

Red bar of Figure 2: after release nothing is cancelled. Had line 2 been rejected by the bank instead, X5 applies.

X5 · Exceptions (class 7): a returned payout, and a reroute with custody

Two of the five sub-classes, with the figures of Patterns F and D.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Bank rejects a GBP payout (deal 6, 2,000 USDT)	SETTLEMENT_RETURN returned_amount 1,513.80, fee_kept 15.00; line outcome returned	client net 1,528.80 (L5); bounce fee 15.00 absorbed	T24: DUE Dr 1,513.80 · PAYABLE Cr 1,513.80 · EXP_BANKFEE Dr 15.00 · POOL GBP Cr 15.00 · DUE Dr 15.00 · PAYABLE Cr 15.00	DUE GBP 1,528.80 again; POOL GBP −15.00 until squared (T24a: 19.43 USDT of NPL's converted)
2. Return to the sender (C6a outcome 3)	DEAL return leg, funding_source returned_settlement; FEE_DECISION markup waived; CONVERSION GBP→USDT at 1.2850; settlement line to the sender_return receiver	1,528.80 × 1.2850 = 1,964.51 USDT, no fee (§16)	T25: PAYABLE GBP Dr 1,528.80 · DUE GBP Cr 1,528.80 · DUE USDT Dr 1,964.51 · PAYABLE USDT Cr 1,964.51; then a line as T23	NPL's 15.60 GBP margin from the first conversion stays in the pool

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
3. Ali unavailable before conversion (deal 3)	REROUTE phase post_collection, outcome via_own_wallet, alternate Jeton, forward_and_recover_dues 30.00; OWN_WALLET AQN-LT-SUB; CUSTODY view shows the balance and its age	—	T12: CL.OWN.USDT.HELD Dr 10,000 · PAYABLE Cr 10,000; T13: CL.JETON.USDT.COLLECTED Dr 10,000 · HELD Cr 10,000 · CO.OWN.USDT.WALLET Dr 30.00 · CO.JETON.USDT.POOL Cr 30.00	HELD 0 after the forward; Jeton owes 30.00 less; then T14 converts at Jeton
4. Ali cannot pay out after converting (2b case 3)	REROUTE phase post_conversion with return_conversion_id, second_conversion_id, rate_difference −79.77, ratified_by Management	Ali back-converts 8,820.00 EUR at 1.1150 = 9,834.30 USDT; Jeton converts at 0.9000 less 1.25 %: delivered 8,740.23	T56, T57, T58 (Pattern O): client net 8,820.00 unchanged; EXP_PARTNER 110.64; VAR_CONVERSION Cr 30.87; POOL at Jeton Cr 79.77	DUE at Jeton 8,820.00; POOL at Jeton −79.77 until squared; POOL at Ali 90.00 untouched

Money that has moved is never rolled back: each step is a new posting, and the client's entitlement is the same figure throughout.

X6 · Balances, retention and offsets (classes 5.2 and 7.4): money that moves without a new deal

Patterns H and N.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Balance conversion	BALANCE_CONVERSION 5,000 SGD → USDT at Ali (Finance requests, Management approves); CONVERSION owned by it; ENTITLEMENT_REATTRIBUTION for Group B	5,000 × 0.7400 = 3,700.00 USDT, no markup (§16)	T27: PAYABLE SGD Dr 5,000 · DUE SGD Cr 5,000 · DUE USDT Dr 3,700.00 · PAYABLE USDT Cr 3,700.00	Group B entitlement now in USDT
2. Earnings offset	EARNINGS_OFFSET from Jeton (2,025.00 EUR) to Ali (2,272.73 USDT at 0.8910); EARNINGS_RECEIVABLE rows at Jeton recovered_by_offset_id	2,025.00 ÷ 0.8910 = 2,272.73 USDT (§13)	T54: DUE Jeton EUR Dr 2,025 · PAYABLE EUR Cr 2,025 · PAYABLE USDT Dr 2,272.73 · COLLECTED Ali Cr 2,272.73 · POOL Ali USDT Dr 2,272.73 · FX_RESULT USDT Cr 2,272.73 · FX_RESULT EUR Dr 2,025 · POOL Jeton EUR Cr 2,025	nothing of NPL's rests at Jeton; NPL's margin now in USDT at Ali
3. Retention at the next conversion	CONVERSION residual_in 1,010.10 USDT on the 100,000 USDT deal at Ali	converted_in = 100,000 − 2,272.73 − 1,010.10 = 96,717.17; delivered 86,175.00 EUR = client net 88,200.00 − 2,025.00 already at Jeton (§13)	T55	POOL at Ali 3,282.83 USDT; client EUR 86,175.00 at Ali + 2,025.00 at Jeton = entitlement 88,200.00 unchanged

The client is neutral throughout; FX_RESULT records NPL's own currency exchange and needs no later closing.

X7 · Shares, rebates and the reseller (classes 5.1, 8.2 and the NPL-GR project)

Patterns J and M.

Step	Record written (ERD)	Figure (Calculation Specification)	Posting (Ledger Posting Design)	Balances after
1. Markup share accrued with a conversion	MARKUP_SHARE_ACCRUAL on deal 3, direction payable, 25 %	67.50 × 25 % = 16.88 EUR (§11)	T30: EARN_GROSS Dr 16.88 · CO.PARTY_P.EUR.SHARE_PAYABLE Cr 16.88; T31 at month end from NPL's wallet	EARN_GROSS net of shares
2. Raeen sub-account and Aquanow rebate	CONVERSION at Aquanow (fee on market 0.80 %); MARKUP_SHARE_ACCRUAL receivable 144.00 (party_retains); PARTNER_REBATE_ACCRUAL expected 30.00 USDT	sender 4 % = 360.00; partner cost 72.00; share base 288.00; Raeen 50 % = 144.00; rebate 0.3 % × 10,000 = 30.00 (L6, §11)	T32: DUE 8,640 · SHARE_RECEIVABLE Dr 144 · EARN_GROSS Dr 144 · EXP_PARTNER Dr 72 · PAYABLE Cr 8,640 · EARN_GROSS Cr 360; T33: REBATE_RECEIVABLE Dr 30 · EARN_REBATE Cr 30; T34/T35 at month end (statement 29.50 → true-up 0.50)	SHARE_RECEIVABLE 144.00 until Raeen pays; EARN_REBATE 29.50 after the true-up
3. NPL-GR: two invoices, one remittance deal	INVOICE vendor 9,000.00 EUR and resale 9,450.00 EUR; DEAL_GROUP target_out 10,852.09 USDT collected as THB cash 360,000 (cash-rounded); vendor paid in two instalments at the day's rates	amount_in 359,746.78 → 360,000 THB; client net 10,859.73 incl. 7.63 surplus; earnings 106.42 (L7); true-ups 0 and 35.19 (§14)	T44 to T51: RESELL_COST/VENDOR_PAYABLE, RESALE_RECEIVABLE/RESELL_REVENUE, the remittance deal as any client, VAR_FX_TIMING on each instalment, the margin line 521.76 USDT to NPL-GR's wallet	reseller revenue 450.00 EUR intact; remittance margin 144.25 USDT = 106.42 + 37.83 forex; NPL_PROFIT = both, per currency

Rebates are NPL's alone and never in a share base; the reseller revenue is fixed when the two invoices are posted and no later transaction touches it.

Closing. after X1 to X4 the identities of the Ledger Posting Design hold at every step (client assets = client liabilities per currency; PAYABLE = Σ GROUP_ENTITLEMENT; the partner's one balance = client balances + NPL's pool), and the reconciliation record of a month end (Pattern Q, T59) proves them against Ali's statement. The opening entries of the cutover (Pattern R, T60, T61) are the same shapes posted once, on the opening date.

Source: NPL's Fee Outlines workbook and NPL's answers of 8–9 October 2026, consolidated into the model.